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CMPUT 365: Introduction to Reinforcement Learning

## Worksheet 8: Planning, Learning & Acting

**Question 1.** An agent observes the following two episodes from an MDP,

$$S_0 = 0, A_0 = 1, R_1 = 1, S_1 = 1, A_1 = 1, R_2 = 1$$

$$S_0 = 0, A_0 = 0, R_1 = 0, S_1 = 0, A_1 = 1, R_2 = 1, S_2 = 1, A_2 = 1, R_3 = 1$$

and updates its deterministic model accordingly. What would the model output for the following queries:

(a) Model($S = 0, A = 0$):

(b) Model($S = 0, A = 1$):

(c) Model($S = 1, A = 0$):

(d) Model($S = 1, A = 1$):

(a) Model($S = 0, A = 0$): 0, 0

(b) Model($S = 0, A = 1$): 1, 1

(c) Model($S = 1, A = 0$): None

(d) Model($S = 1, A = 1$): 1, `terminal`

**Question 2.** An agent is in a 4-state MDP, $\mathcal{S} = \{1, 2, 3, 4\}$, where each state has two actions $\mathcal{A} = \{1, 2\}$. Assume the agent saw the following trajectory,

$$\begin{aligned}
&S_0 = 1, A_0 = 2, R_1 = -1,\\
&S_1 = 1, A_1 = 1, R_2 = 1,\\
&S_2 = 2, A_2 = 2, R_3 = -1,\\
&S_3 = 2, A_3 = 1, R_4 = 1,\\
&S_4 = 3, A_4 = 1, R_5 = 100,\\
&S_5 = 4
\end{aligned}$$

and uses Tabular Dyna-Q with 5 planning steps for each interaction with the environment.

(a) Once the agent sees $S_5$, how many Q-learning updates has it done with **real experience**? How many updates has it done with **simulated experience**?

(b) Which of the following are possible (or not possible) simulated transitions $\{S, A, R, S'\}$ given the above observed trajectory with a deterministic model and random search control?

(i) $\{S = 1, A = 1, R = 1, S' = 2\}$

(ii) $\{S = 2, A = 1, R = -1, S' = 3\}$

(iii) $\{S = 2, A = 2, R = -1, S' = 2\}$

(iv) $\{S = 1, A = 2, R = -1, S' = 1\}$

(v) $\{S = 3, A = 1, R = 100, S' = 5\}$

(a) 5 updates with real experience and 25 updates with simulated experience.

(b) See answers below.

(i) Possible

(ii) Not possible

(iii) Possible

(iv) Possible

(v) Not possible

**Question 3.** Modify the Tabular Dyna-Q algorithm so that it uses Expected Sarsa instead of Q-learning. Assume that the target policy is $\epsilon$-greedy. What should we call this algorithm?

**Tabular Dyna-Q**

Initialize $Q(s,a)$ and $Model(s,a)$ for all $s \in \mathcal{S}$ and $a \in \mathcal{A}(s)$
Loop forever:
(a) $S \leftarrow$ current (nonterminal) state
(b) $A \leftarrow \varepsilon$-greedy$(S,Q)$
(c) Take action $A$; observe resultant reward, $R$, and state, $S'$
(d) $Q(S,A) \leftarrow Q(S,A) + \alpha\big[R + \gamma \max_a Q(S',a) - Q(S,A)\big]$
(e) $Model(S,A) \leftarrow R, S'$ (assuming deterministic environment)
(f) Loop repeat $n$ times:
$S \leftarrow$ random previously observed state
$A \leftarrow$ random action previously taken in $S$
$R, S' \leftarrow Model(S,A)$
$Q(S,A) \leftarrow Q(S,A) + \alpha\big[R + \gamma \max_a Q(S',a) - Q(S,A)\big]$

To make the algorithm use Expected Sarsa instead of Q-learning, we should change the updates made using both real and simulated experience, as shown below:

$$Q(S,A) \leftarrow Q(S,A) + \alpha\Big[R + \gamma \sum_a \pi(a|S')Q(S',a) - Q(S,A)\Big].$$

To make the target policy $\epsilon$-greedy with respect to the estimated $Q$ values, we can use the following definition. For all states $s \in \mathcal{S}$ and actions $a \in mathscrA$,

$$\pi(a|s) = \begin{cases} 1 - \epsilon + \frac{\epsilon}{|A|}, \text{ if } a = \arg\max_{a'} Q(s,a') \\ \frac{\epsilon}{|A|}, \text{ otherwise.} \end{cases}$$

We could name this algorithm as Dyna-Expected-Sarsa.

**Question 4.** Consider an MDP with two states $\{1,2\}$ and two possible actions: $\{\text{stay}, \text{switch}\}$. The state transitions are deterministic, the state does not change if the action is "stay" and the state switches if the action is "switch". However, rewards are randomly distributed:

$$P(R=r|S=1,A=\text{stay}) = \begin{cases} 0.4 & \text{for } r=0 \\ 0.6 & \text{for } r=1 \end{cases}, \quad P(R=r|S=1,A=\text{switch}) = \begin{cases} 0.5 & \text{for } r=0 \\ 0.5 & \text{for } r=1 \end{cases}$$

$$P(R=r|S=2,A=\text{stay}) = \begin{cases} 0.6 & \text{for } r=0 \\ 0.4 & \text{for } r=1 \end{cases}, \quad P(R=r|S=2,A=\text{switch}) = \begin{cases} 0.5 & \text{for } r=0 \\ 0.5 & \text{for } r=1 \end{cases}$$

(a) How might you learn the reward model? Hint: think about how probabilities are estimated. For example, what if you were to estimate the probability of a coin landing on heads? If you observed 10 coin flips with 8 heads and 2 tails, then you can estimate the probabilities by counting: $p(\text{heads}) = \frac{8}{10} = 0.8$ and $p(\text{tails}) = \frac{2}{10} = 0.2$.

(b) Modify the tabular Dyna-Q algorithm to handle this MDP with stochastic rewards.

(a) We can estimate the reward distribution by keeping track of how many times the tuples $(s, a, r)$ and $(s, a)$ occur for each state $s$, action $a$, and reward $r$. In particular, let $\hat{p}(r|s, a)$ be an estimate of $Pr(R = r|S = s, A = a)$. Then

$$\hat{p}(r|s, a) = \frac{\text{count}(s, a, r)}{\text{count}(s, a)}.$$

(We do not explicitly need to maintain the count of $(s, a)$ pairs; it can be computed as $\text{count}(s, a) = \sum_{r \in \mathcal{R}} \text{count}(s, a, r)$.)

(b) At the beginning of the algorithm, we initialize counts of all the $(s, a, r)$ and $(s, a)$ tuples to zero:

$$\text{count}(s, a, r) = \text{count}(s, a) = 0, \qquad \text{for all} \quad s, a, r \in \mathcal{S} \times \mathcal{A} \times \mathcal{R},$$

and initialize the estimate of the reward model, say, uniformly as follows

$$\hat{p}(r|s, a) = 1/|\mathcal{R}|, \qquad \text{for all} \quad s, a, r \in \mathcal{S} \times \mathcal{A} \times \mathcal{R}.$$

Now the model $\text{Model}(S, A)$ would not store the next deterministic state and reward, but rather the next state (which is still deterministic) and the reward distribution $\hat{p}$. In particular, line (e) in the algorithm Tabular Dyna-Q would now consist of the following steps:

(i) Update the counts: $\text{count}(S, A) = \text{count}(S, A) + 1$ and $\text{count}(S, A, R) = \text{count}(S, A, R) + 1$.

(ii) Estimate the reward distribution: $\hat{p}(R|S, A) = \frac{\text{count}(S,A,R)}{\text{count}(S,A)}$.

(iii) Update the model: $\text{Model}(S, A) = \hat{p}(\cdot|S, A), S'$.

Finally, during the planning from simulated experience step, we will sample the reward using this estimated reward distribution and use the state stored in the model, that is $\tilde{p}, S' = \text{Model}(S, A)$ and $R \sim \tilde{p}(\cdot)$. The complete algorithm is presented below.

**Dyna-$Q$ with Stochastic Rewards**

Initialize $Q(s, a)$ and $\text{Model}(s, a)$, for all $s, a \in \mathcal{S}, \mathcal{A}$
Initialize $\text{count}(s, a, r) = \text{count}(s, a) = 0$, for all $s, a, r \in \mathcal{S} \times \mathcal{A} \times \mathcal{R}$
Initialize $\hat{p}(r|s, a) = 1/|\mathcal{R}|$, for all $s, a, r \in \mathcal{S} \times \mathcal{A} \times \mathcal{R}$
Loop forever:
- (a) $S \leftarrow$ current (non-terminal) state
- (b) $A \leftarrow \epsilon\text{-greedy}(S, Q)$
- (c) Take action $A$; observe the resulting reward $R$ and state $S'$
- (d) $Q(S, A) \leftarrow Q(S, A) + \alpha[R + \gamma \max_a Q(S', a) - Q(S, A)]$
- (e1) $\text{count}(S, A) = \text{count}(S, A) + 1$ and $\text{count}(S, A, R) = \text{count}(S, A, R) + 1$
- (e2) $\hat{p}(R|S, A) = \frac{\text{count}(S,A,R)}{\text{count}(S,A)}$
- (e3) $\text{Model}(S, A) \leftarrow \hat{p}(\cdot|S, A), S'$
- (f) Loop repeat $n$ times:
  - $S \leftarrow$ random previously observed state
  - $A \leftarrow$ random action previously taken in $S$
  - $\tilde{p}, S' \leftarrow \text{Model}(S, A)$
  - Sample $R \sim \tilde{p}(\cdot)$
  - $Q(S, A) \leftarrow Q(S, A) + \alpha[R + \gamma \max_a Q(S', a) - Q(S, A)]$

**Question 5.** Consider an MDP with three states $\mathcal{S} = \{1, 2, 3\}$, where each state has two possible actions $\mathcal{A} = \{1, 2\}$ and a discount rate $\gamma = 0.5$. Suppose estimates of $Q(S, A)$ are initialized to 0 and you observed the following episode according to an unknown behaviour policy where $S_3$ is the terminal state.

$$S_0 = 1, A_0 = 1, R_1 = -7, S_1 = 2, A_1 = 2, R_2 = 5, S_2 = 1, A_2 = 1, R_3 = 10$$

(a) Suppose you used Q-learning with the above trajectory to estimate $Q(S, A)$, what are your new estimates for $Q(S = 1, A = 1)$ using $\alpha = 0.1$?

(b) What is one possible model for this environment? Is the model stochastic or deterministic?

(c) Suppose in the planning loop, after search control, we would like to update $Q(S = 1, A = 1)$ with Q-planning. What are the possible outputs of Model$(S = 1, A = 1)$?

(d) If your model outputs $R = R_3$ and $S' = S_3$, what is $Q(S = 1, A = 1)$ after one Q-planning update? Use the estimates of $Q(S, A)$ from before.

(a) The update for the first transition:

$$Q(1, 1) \leftarrow Q(1, 1) + 0.1 \cdot [-7 + 0.5 \max_a Q(2, a) - Q(1, 1) = -0.7]$$

The update for the second transition:

$$Q(1, 1) \leftarrow Q(1, 1) + 0.1 \cdot [10 + 0.5 \cdot 0 - Q(1, 1) = -0.7 + 1 + 0.07 = 0.37]$$

(b) Any model that is consistent with the trajectory. The probability for the transitions in the trajectory should not be zero. The model is stochastic since in state 1 with action 1, we can transition to both state 2 or the terminal state.

(c) The possible outputs of the model are (-7,2) and (10, `terminal`).

(d)

$$Q(1, 1) \leftarrow Q(1, 1) + 0.1 \cdot [10 + 0.5 \cdot 0 - Q(1, 1)] = 0.37 + 1 - 0.037 = 1.333$$

**Question 6.** Exercise 8.2 from Sutton & Barto (2018).

Dr. Sutton has asked that the solutions to the textbook exercises not be made electronically available to students. You are encouraged to attend office hours, where the teaching assistants can discuss the exercises with you and show you the answers. Alternatively, if you email Dr. Sutton your answers for an entire chapter, he will likely send you the electronic answer sheet for that chapter.

**Question 7.** Exercise 8.3 from Sutton & Barto (2018).

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